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Zipper Entanglement Renormalization on One Dimensional Free Fermion Systems

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Question

How do we obtain an effective description for a free fermion ground state?

Two Approaches

Hamiltonian + Ansatz → **States**

(Tensor Network)

Tensor Network Description ← **States (Given in other format)**

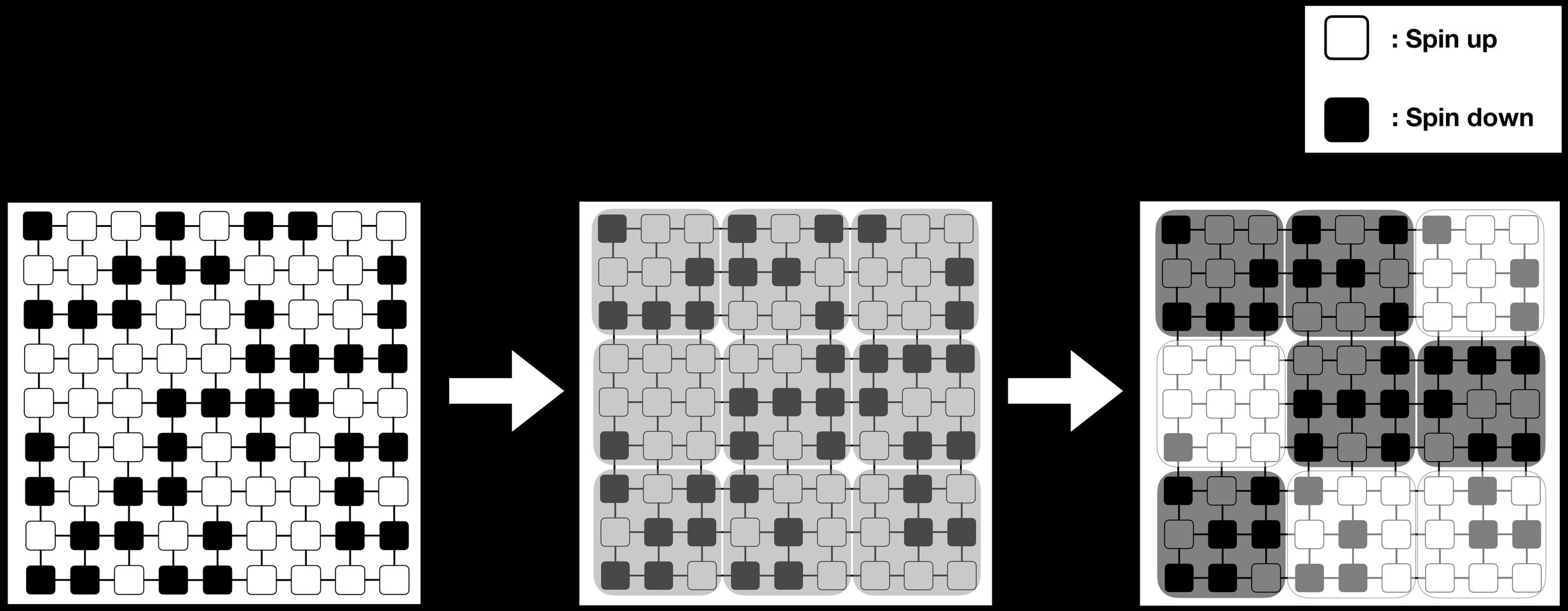
Effectiveness ?

Effectiveness

Tensor Network States

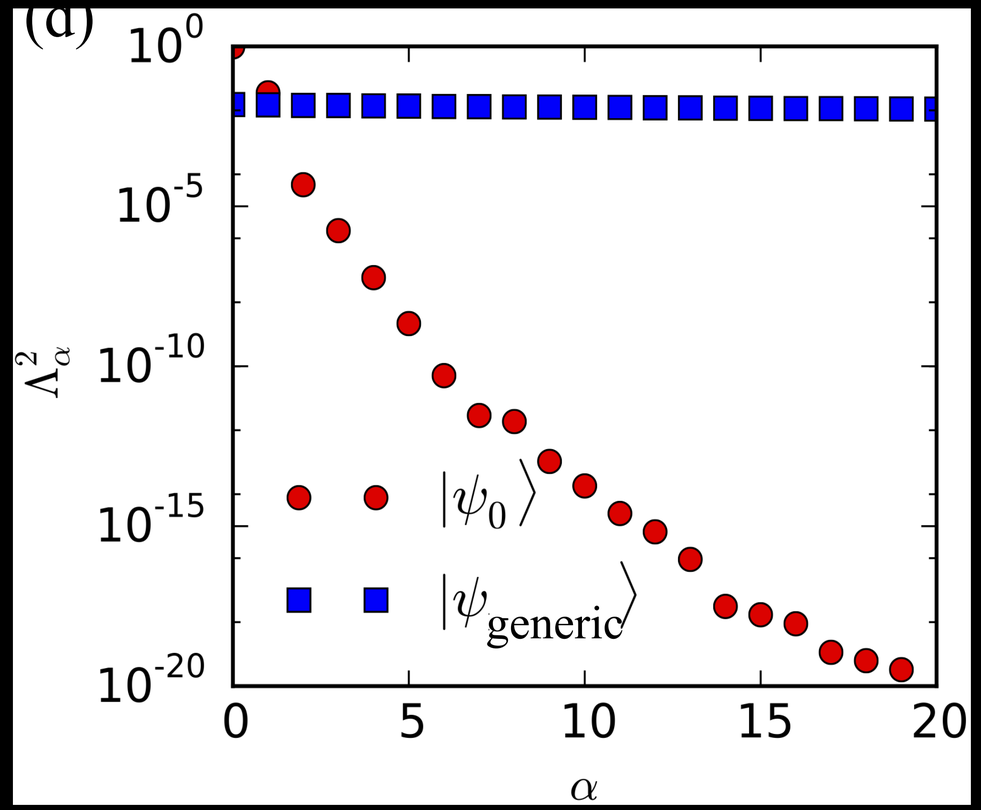
Renormalization Group

Renormalization Group - Coarse-Graining



Coarse-Graining Transformation of Ising Model

Quantum Entanglement



[Hauschild & Pollmann (2018)]

$$S = -\,\mathbf{Tr}(\rho_R \ln \rho_R)$$
$$= -\sum_\alpha \Lambda_\alpha^2 \ln \Lambda_\alpha^2$$

$$0 \leq \Lambda_\alpha^2 \leq 1$$
$$\Lambda_\alpha^2 = Eig(\rho_R)$$

Volume Law

$S \sim Vol(A)$

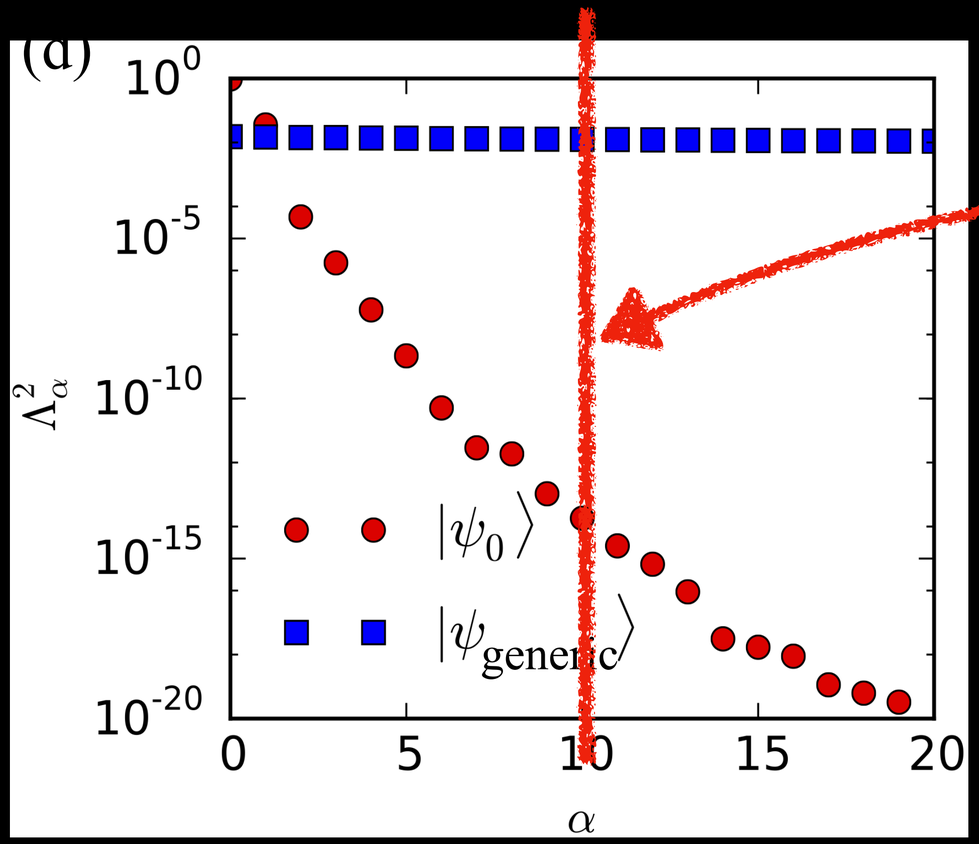
Area Law

$S \sim \partial A$

[Hasting (2007)]

Truncation as Discarding Irrelevant Information

- Discard modes with low entanglement contribution



Truncation

(e.g. $\chi = \alpha = 10$)

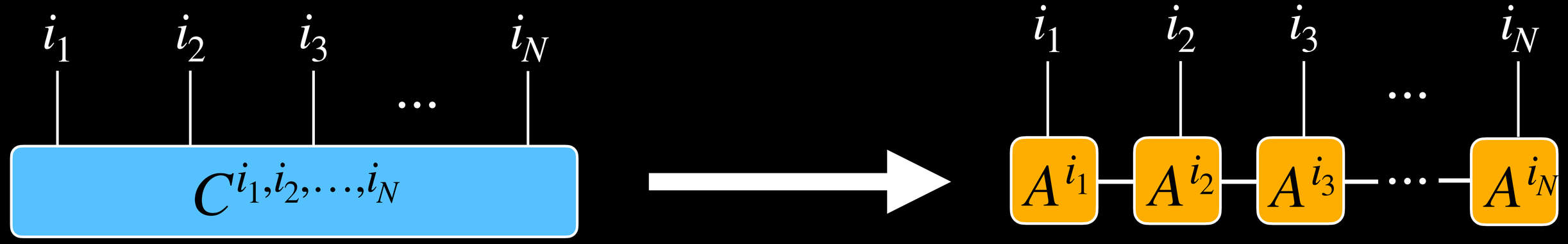
Effective Description

(Use less degrees of freedom)

[Hauschild & Pollmann (2018)]

Matrix Product States (MPS)

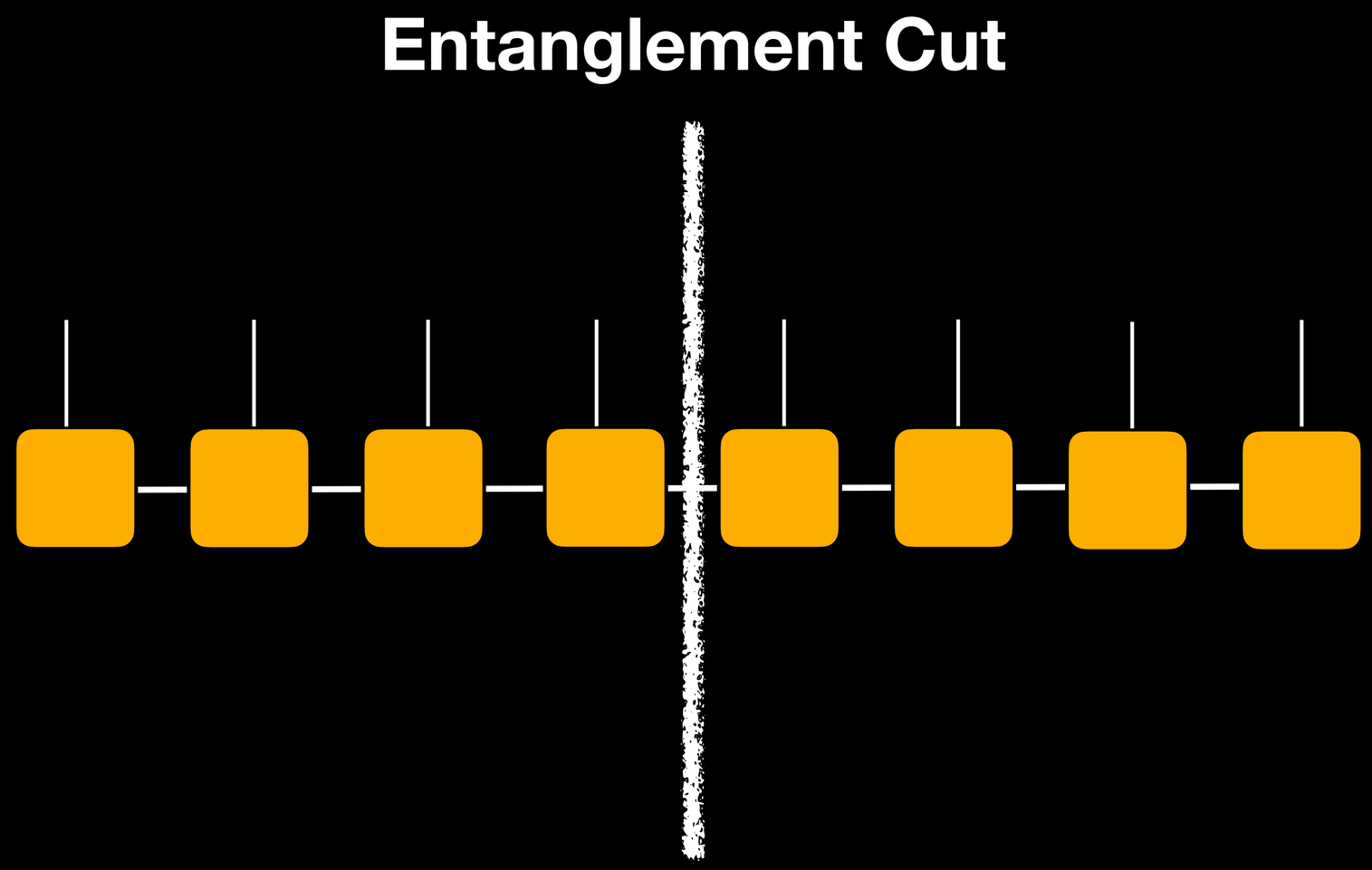
$$|\Psi\rangle = \sum_{\{i\}} C^{i_1,i_2,\ldots,i_N} |i_1\rangle |i_2\rangle \cdots |i_N\rangle \qquad |\Psi\rangle \approx \sum_{\{i\}} (A^{i_1} A^{i_2} \cdots A^{i_N})_{\{k\}} |i_1\rangle |i_2\rangle \cdots |i_N\rangle$$



Exp. Large Tensor
(Dof ~ d^N)

Matrices with Smaller size
(Dof ~ $N\chi^2$)

Matrix Product States - Area Law



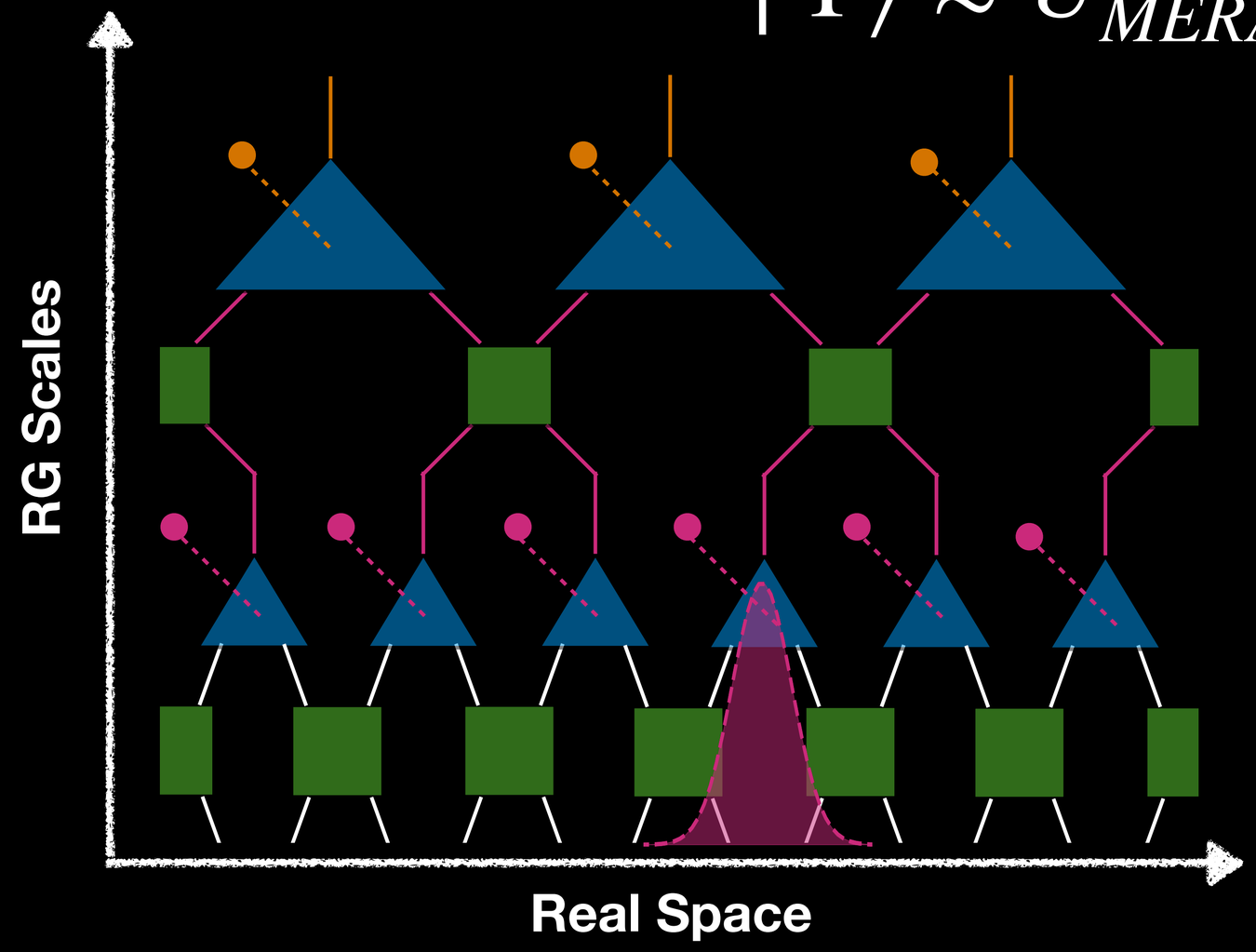
Entanglement Contribution comes from the boundary

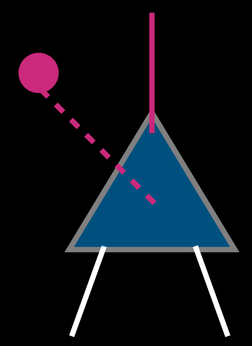
↓

MPS by design satisfies area-law

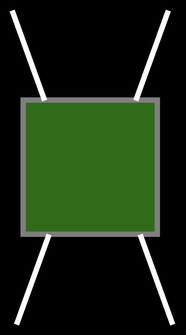
Multiscale Entanglement Renormalization Ansatz (MERA)

$$|\Psi\rangle \approx \hat{U}_{MERA}|0\rangle^{\otimes N}$$



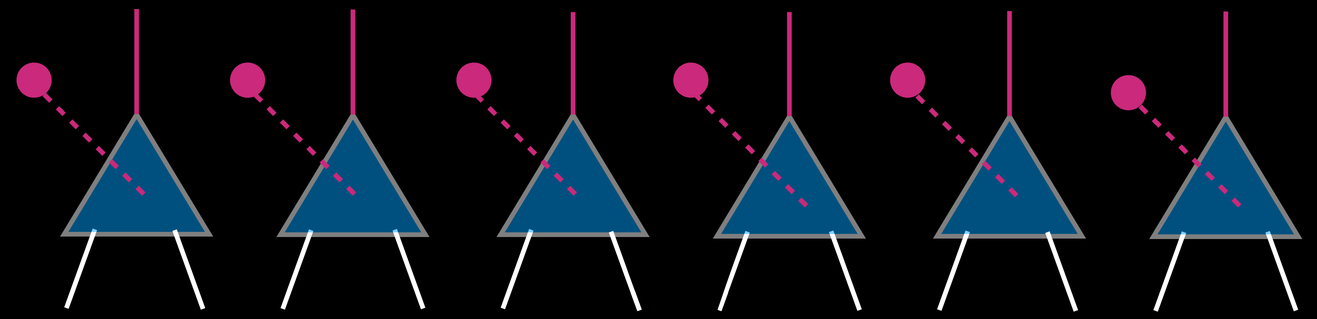


Isometry

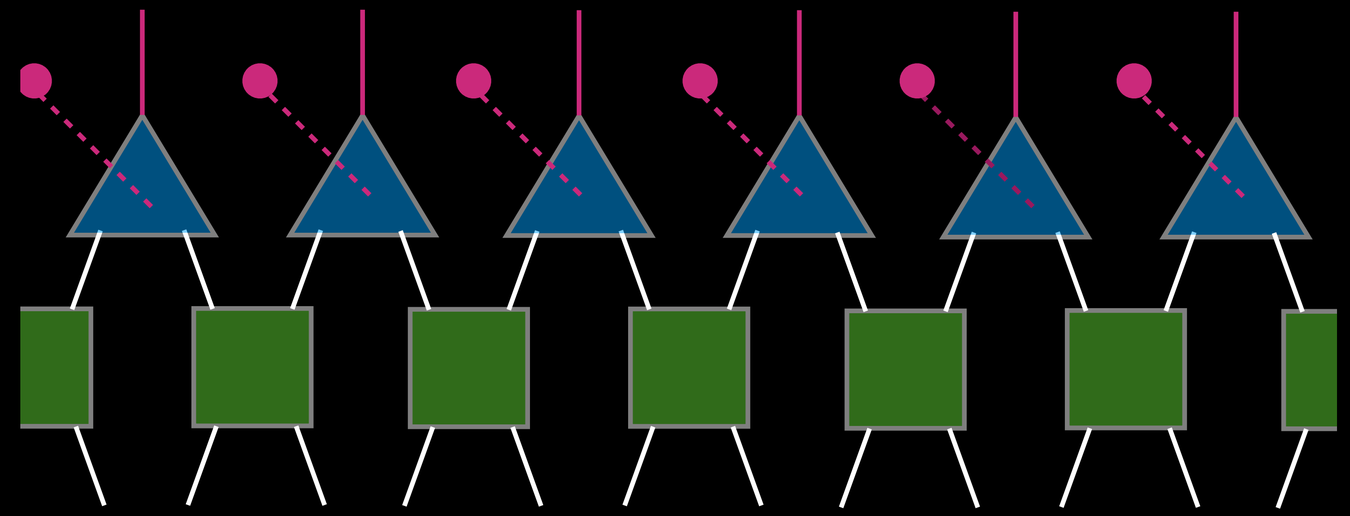


Disentangler

Disentanglers and Coarse-Graining Transformation

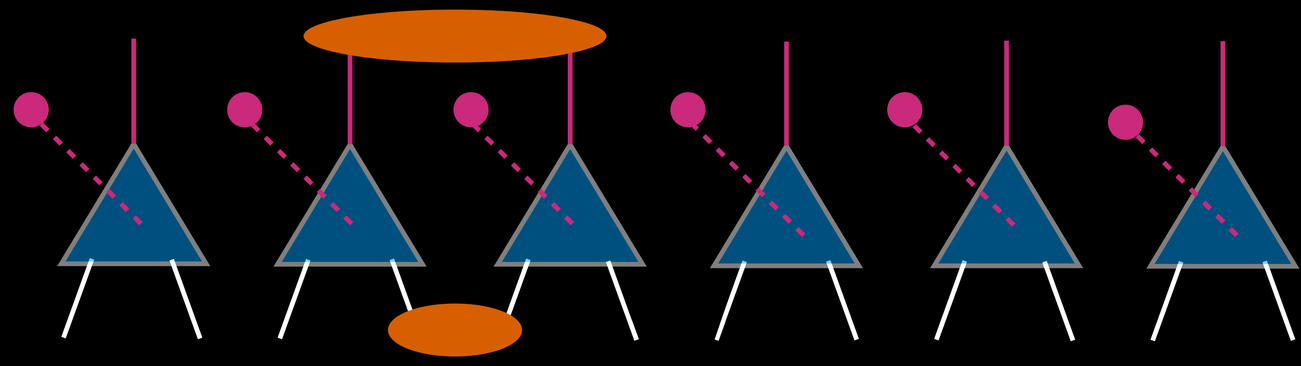


vs.



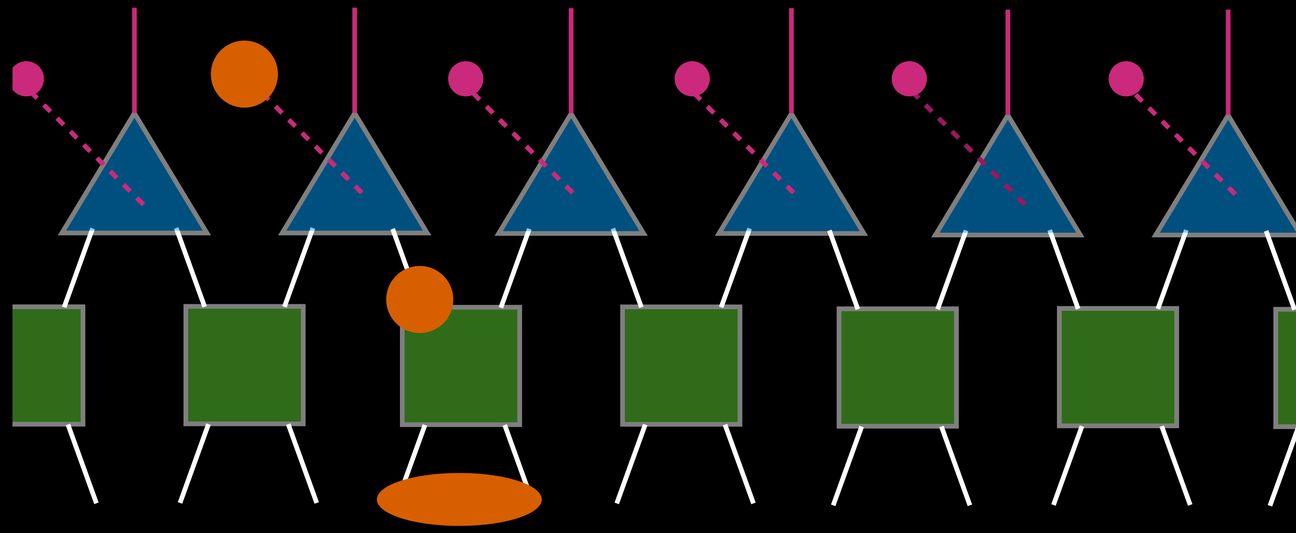
Disentanglers and Coarse-Graining Transformation

Unable to remove Short-Range Entanglement (SRE)

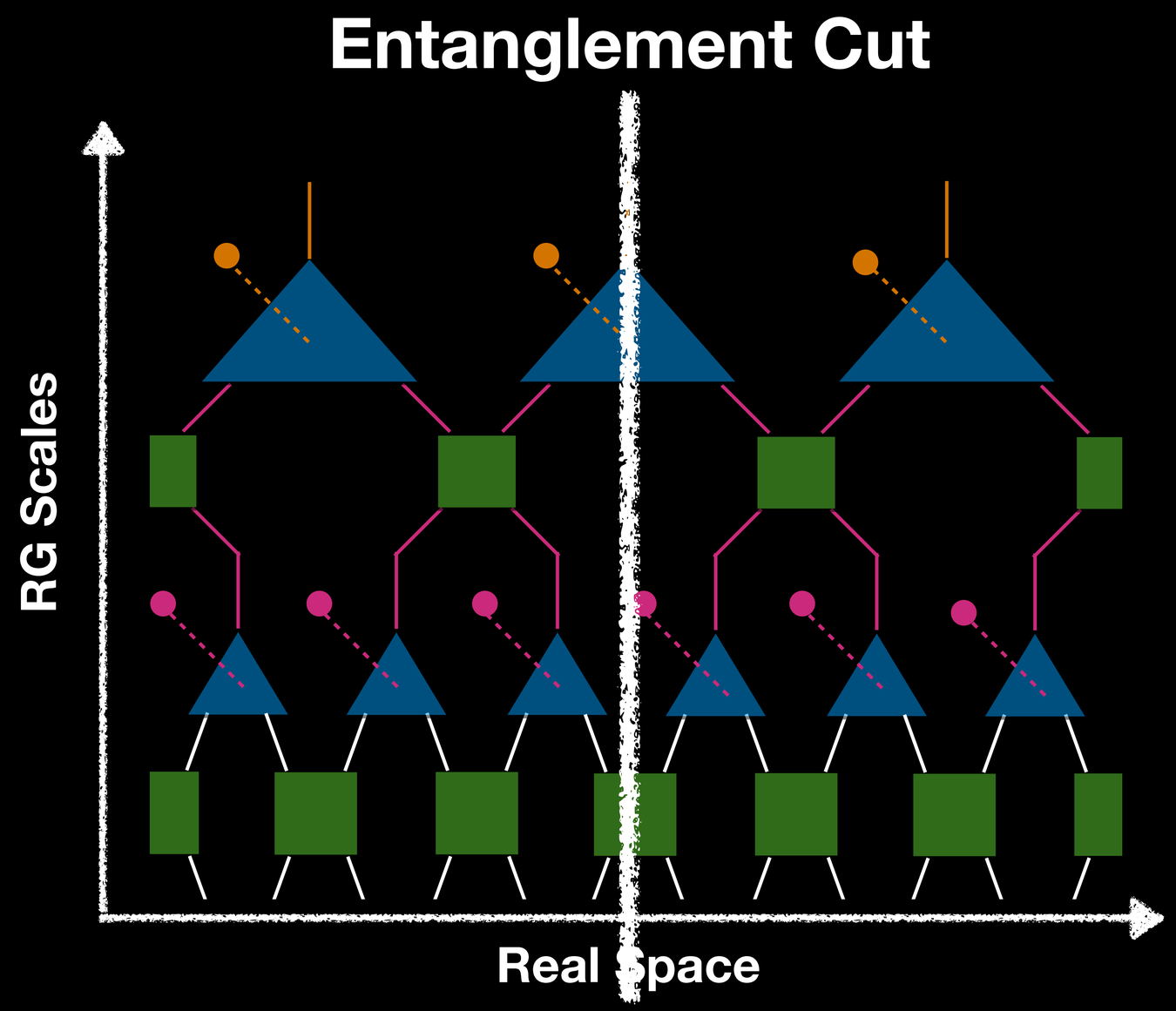


VS.

Able to remove SRE



MERA - Sub-volume law scaling



Each layer ~ MPS

$$S \leq const$$

$$\times$$

Number of Layers

$$\sim \log N$$

$$=$$

$$S \propto \log N$$

Free Fermion Systems/States

- Ground state of Free Fermion Hamiltonian (quadratic Hamiltonian with fermionic bilinear)
  - Single-particle sector provides necessary information
  - Determined by its 2-point correlation function (Wick's Theorem)
  - The entanglement property is computable from the correlation function [Chung & Peschel (2003)]

Entanglement Property is completely known !

Density Matrix, Free Fermion State, and Correlation Matrix

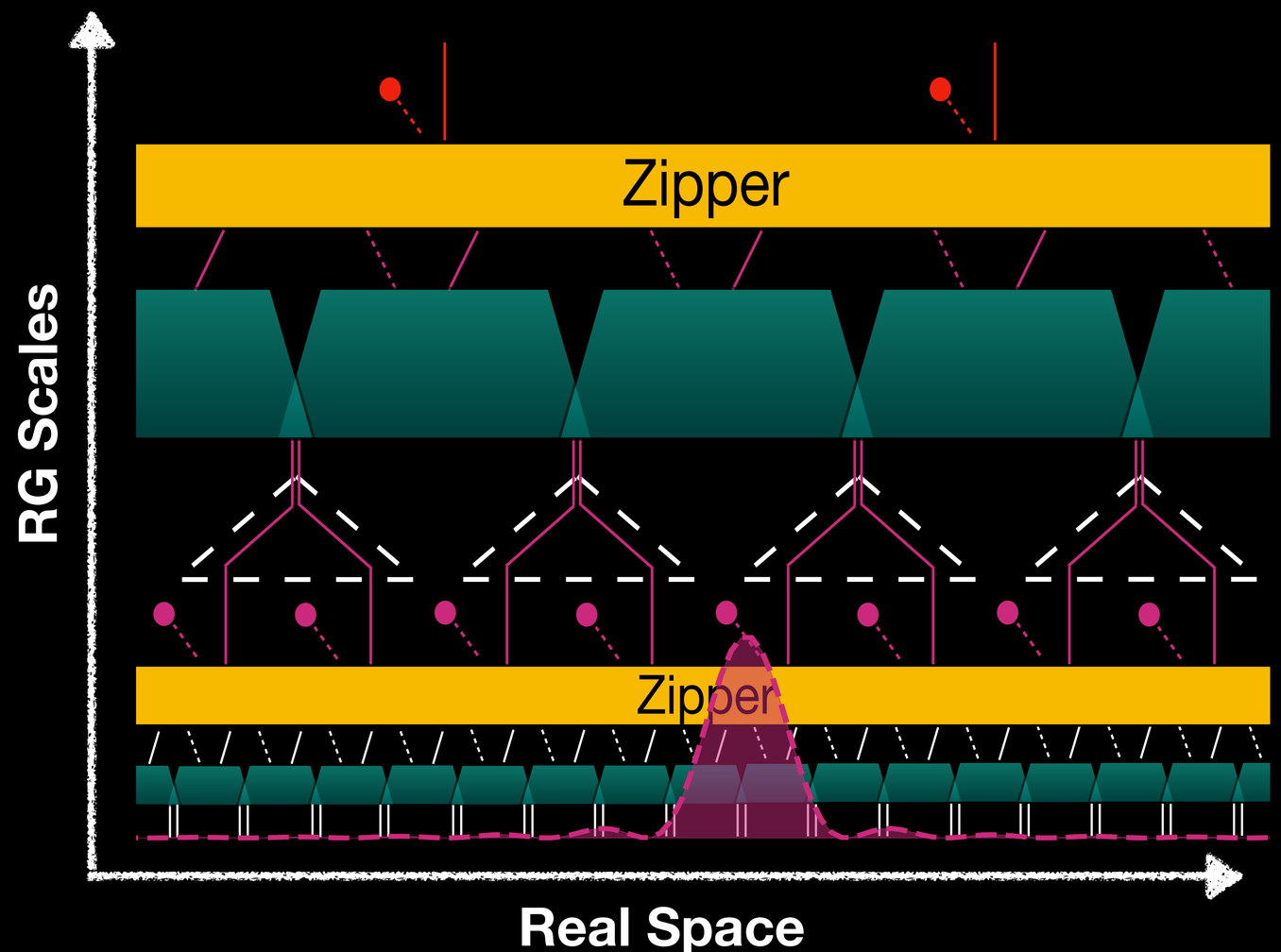
$$|\Psi\rangle \implies \hat{\rho} = |\Psi\rangle\langle\Psi| \iff C_{ij} = \langle\Psi|\hat{c}_i^\dagger \hat{c}_j|\Psi\rangle$$

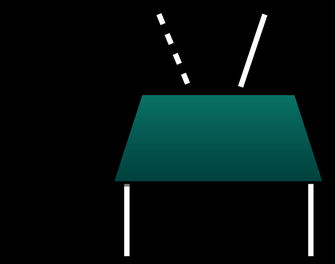
Restriction $\downarrow$ $\qquad\qquad$ $\downarrow$ **Restriction**

$$\hat{\rho}_R \iff (C_R)_{ij},\ i,j \in R$$

Zipper Entanglement Renormalization

$$\hat{U}_{ZER}|\Psi\rangle \approx |\Psi_{core}\rangle \otimes \left(|\Psi_e^{[k]}\rangle \otimes \cdots \otimes |\Psi_e^{[1]}\rangle\right) \otimes \left(|\Psi_f^{[k]}\rangle \otimes \cdots \otimes |\Psi_f^{[1]}\rangle\right)$$





Zipper Unitary

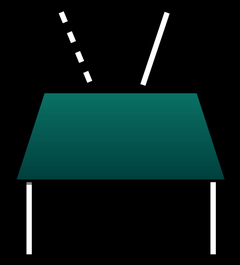
Local Distiller

[Wong, Pang, & Po, 2206.11761]

Overview of ZER Approach

- Construct a coarse-graining transformation for free fermion states
  - Distillers: Identify the short-range and long-range degrees of freedom
    - Frozen Modes (SRE) VS. Courier Modes (LRE)
- Zipper Unitary: Discard short-range entanglement
  - Find a basis transformation for the correlation matrix
  - Decompose it to the contribution of frozen modes and courier modes

Local Distillers



Local Distiller

- Goal: Identify the SRE modes inside a subregion R
- Inspect the contribution of entanglement for different modes inside R
  - Low contribution to the entanglement $\Longrightarrow$ SRE modes (Nearly uncorrelated with the rest)
- Method: Inspect the restricted correlation matrix for a subregion

$$\rho_R \Longleftrightarrow (C_R)_{ij},\ i,j \in R$$

Local Distiller

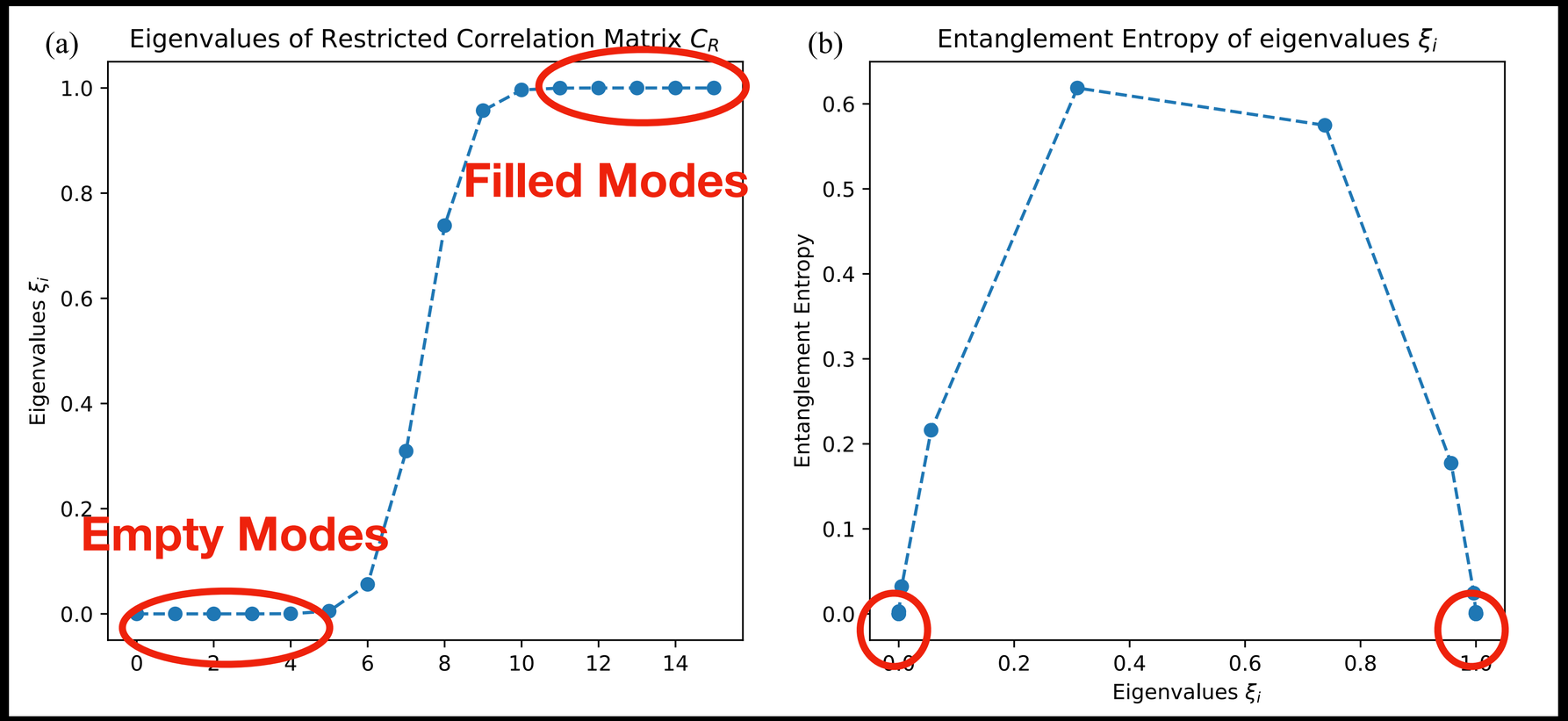
- SRE modes (Frozen Modes)
  - Filled Modes: $\xi \geq 1 - \varepsilon$
  - Empty Modes: $\xi \leq \varepsilon$

$$\xi = \textbf{Eig}(C_R) \; ; \; 0 \leq \xi \leq 1$$

- Local distiller = Projector to the space spanned by filled/ empty modes
- $|\phi_{e/f}\rangle$ is the empty/ filled modes are the eigenvectors of C_R

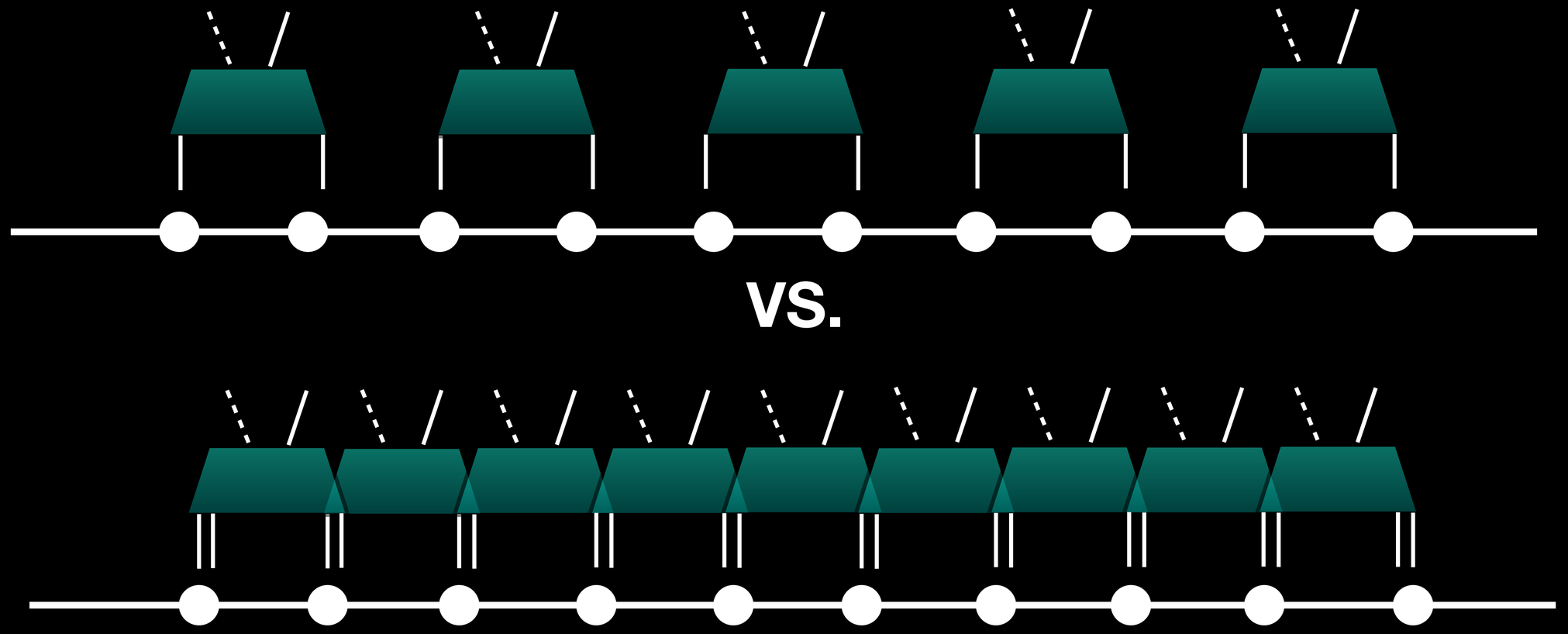
$$h^R_{distill} = \sum_i |\phi_e\rangle_i \langle\phi_e|_i - \sum_j |\phi_f\rangle_j \langle\phi_f|_j$$

Local Distiller - Filled/Empty Modes Example

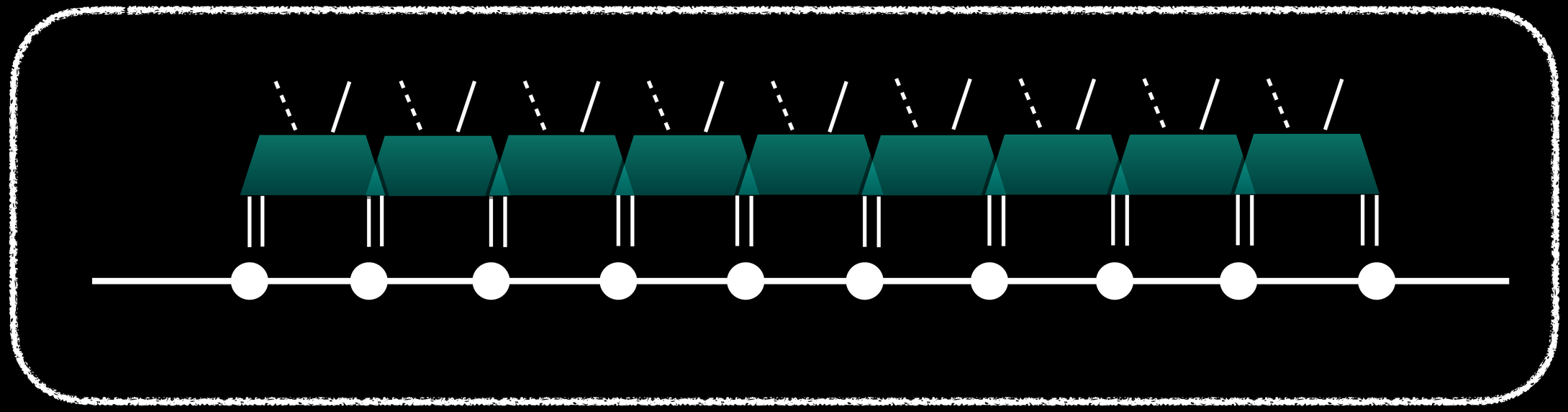


Frozen Modes = Filled + Empty Modes
Nearly disentangled Dofs

Local Distiller - Overlapping Scheme

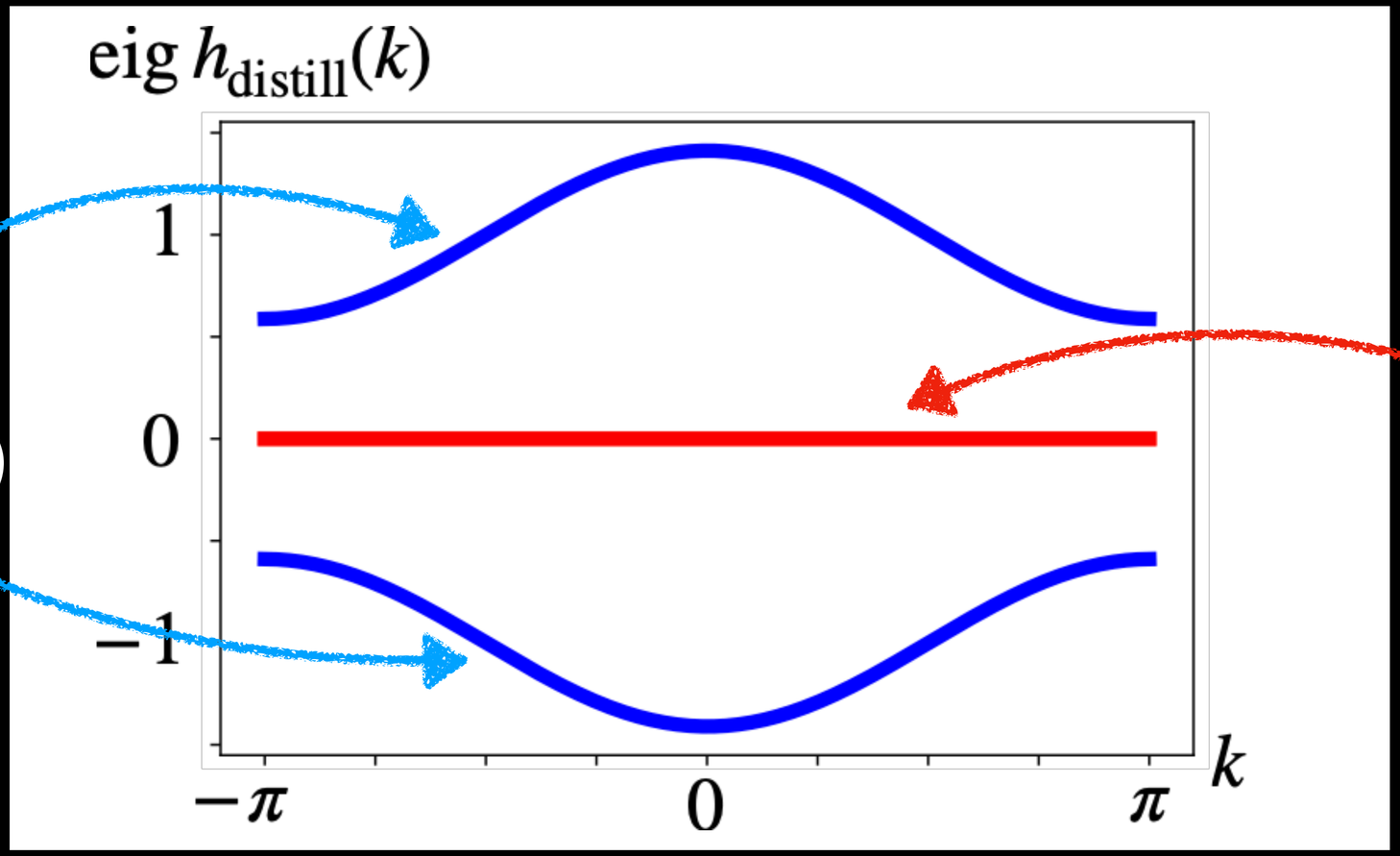


Local Distiller - Overlapping Scheme



Global Distiller = $\sum$ Local Distillers

Global Distiller Spectrum and Courier Modes



Frozen Modes
(Empty / Filled)

Courier Modes

[Wong, Pang, & Po, 2206.11761]

Zipper Unitary Transformation

$$\hat{U}_{Zipper}|\Psi\rangle \approx |\Psi_{courier}\rangle \otimes |\Psi_{filled}\rangle \otimes |\Psi_{empty}\rangle$$

Zipper

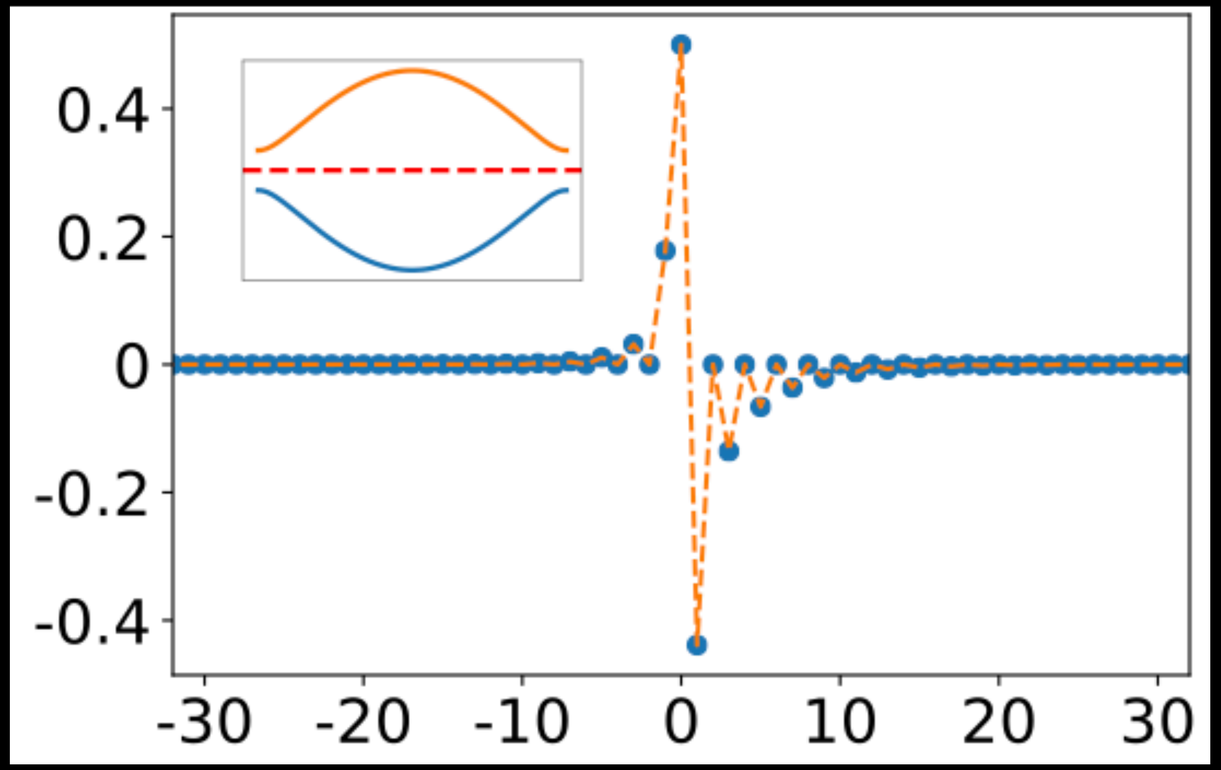
Zipper Unitary

- Zipper Unitary is obtained by Wannierization in real space
  $u_{zipper} = (\psi_{filled}|\psi_{courier}|\psi_{empty})$
- Each Wannier function is quasi-localized in real space (Exp. Decaying tail)
- Rotated correlation matrix

$$C'^T = u^\dagger_{zipper} C^T u_{zipper} \approx \begin{pmatrix} 1 & & \\ & C^T_{courier} & \\ & & 0 \end{pmatrix}$$

- Block-diagonal form of $C_{ij} \Longrightarrow$ Factorization of $|\Psi\rangle$

Su-Schrieffer–Heeger (SSH) Model



Comparison of the original SSH correlation function and the ZER approximation

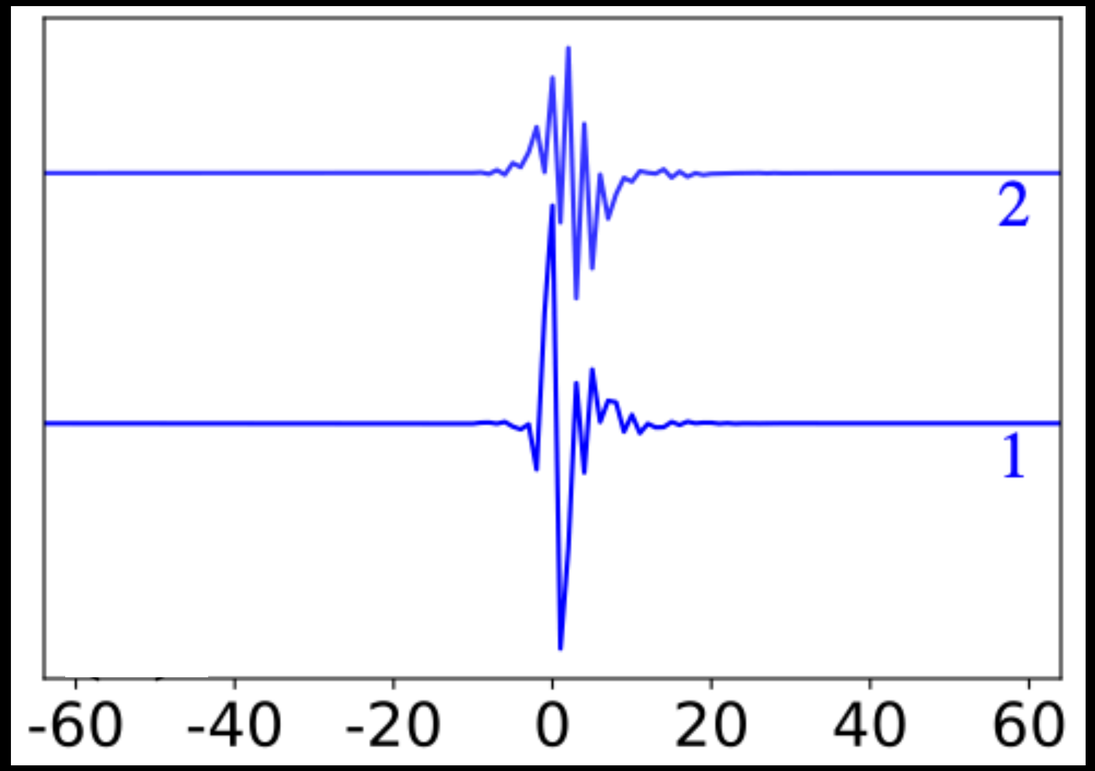
$$C_{ij} = \langle \Psi | \hat{c}_i^\dagger \hat{c}_j | \Psi \rangle$$

vs.

$$C'_{ij} = U_{ZER}^\dagger C_{ZER} U_{ZER}$$

[Wong, Pang, & Po, 2206.11761]

Su-Schrieffer–Heeger (SSH) Model

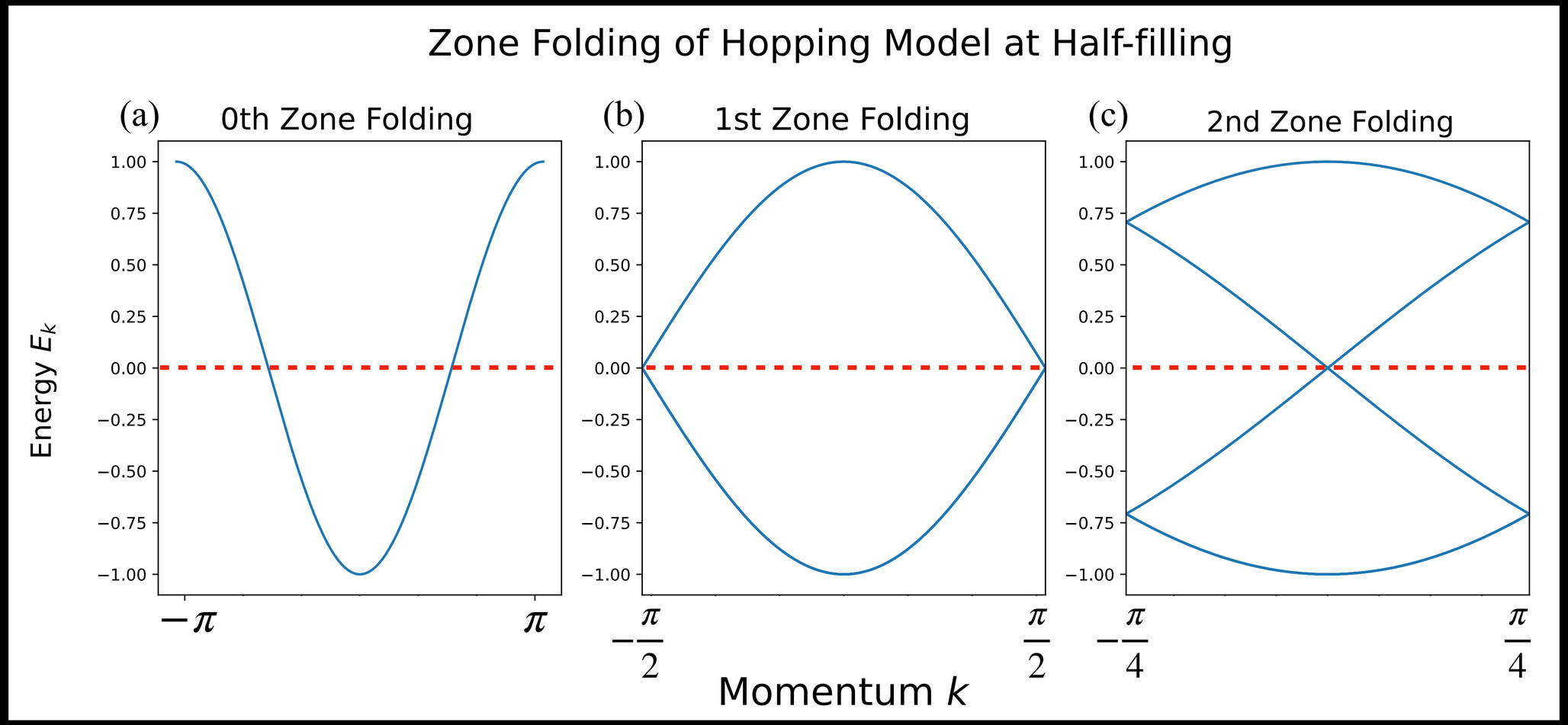


Contribution of Correlation function in different RG times

RG terminates quickly

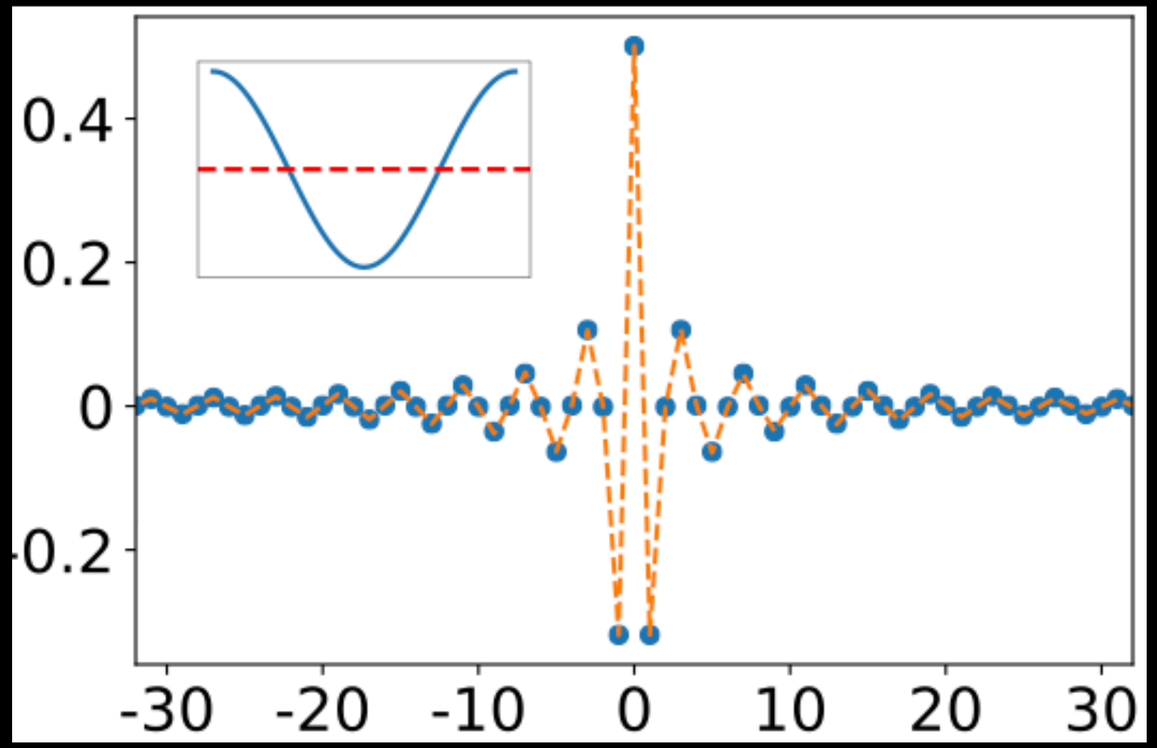
$\Longrightarrow$ MPS description

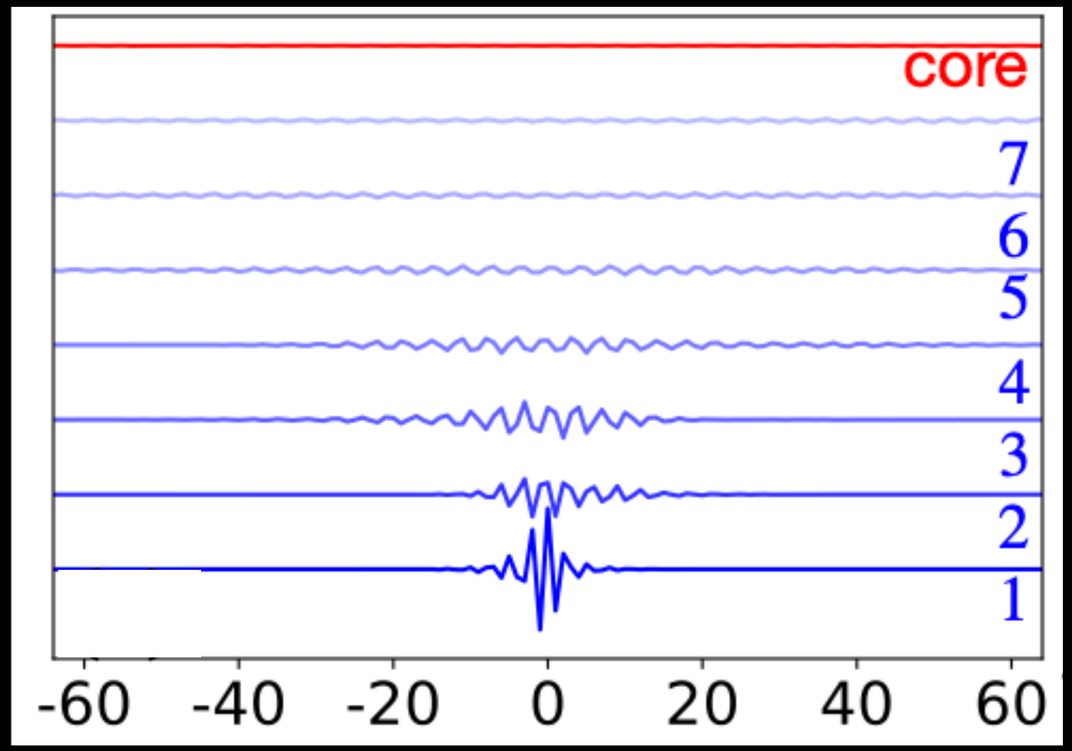
Nearest-Neighbor Hopping Model at Half-filling



Band structure looks the same in further steps $\Longrightarrow$ scale-invariant

Nearest-Neighbor Hopping Model at Half-filling

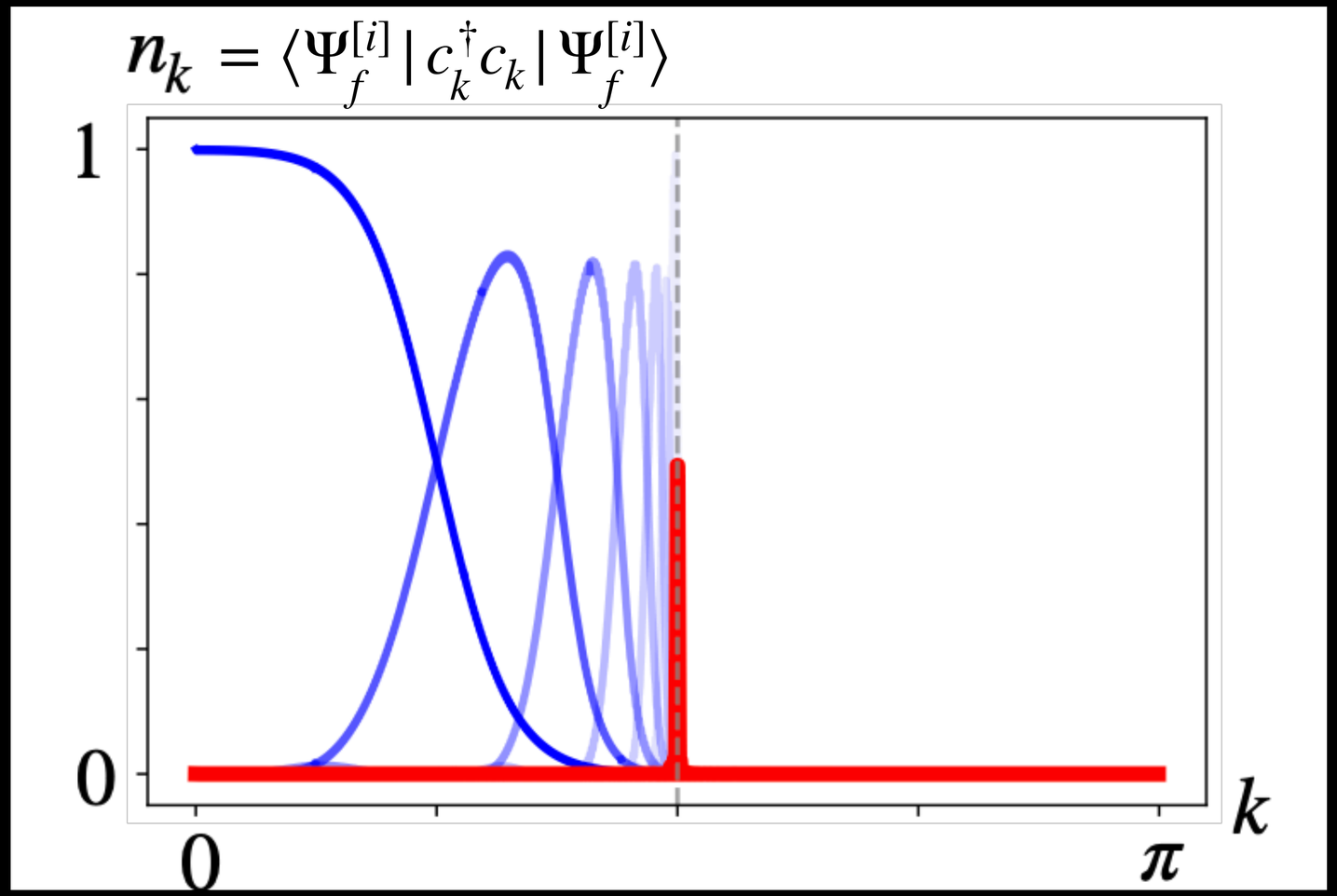




RG terminates not that fast $\Longrightarrow$ MERA description

[Wong, Pang, & Po, 2206.11761]

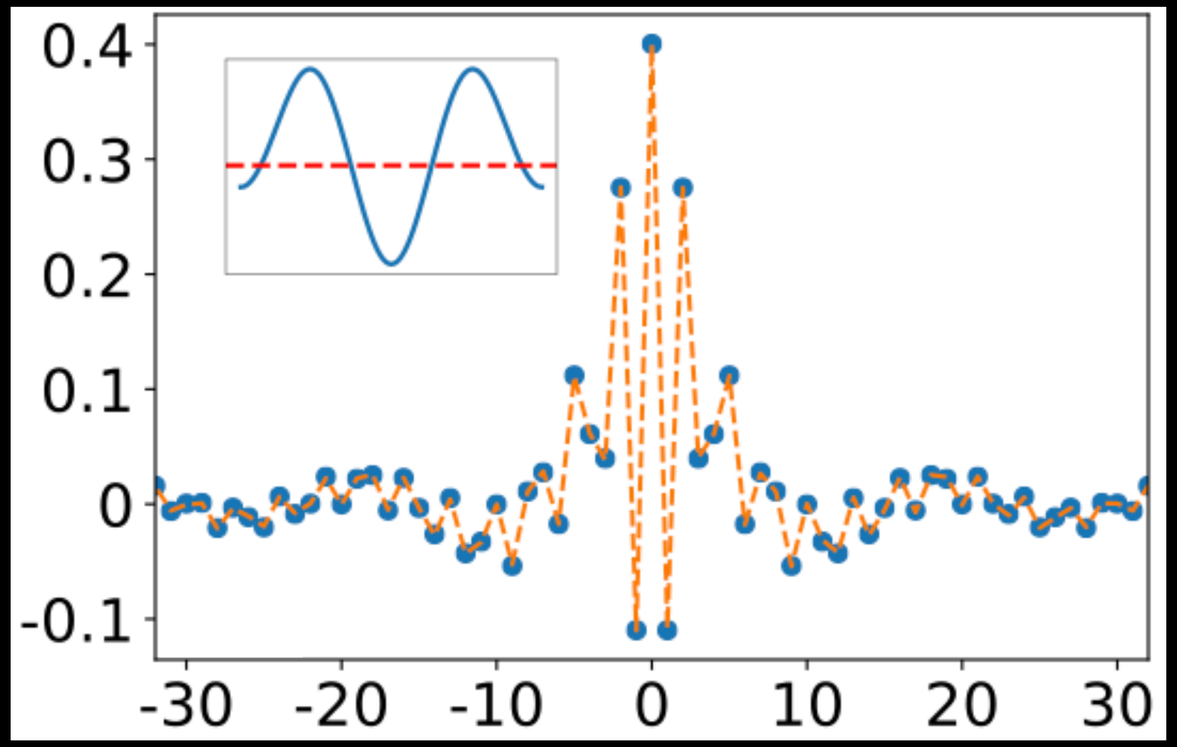
Number occupation n_k at different RG times

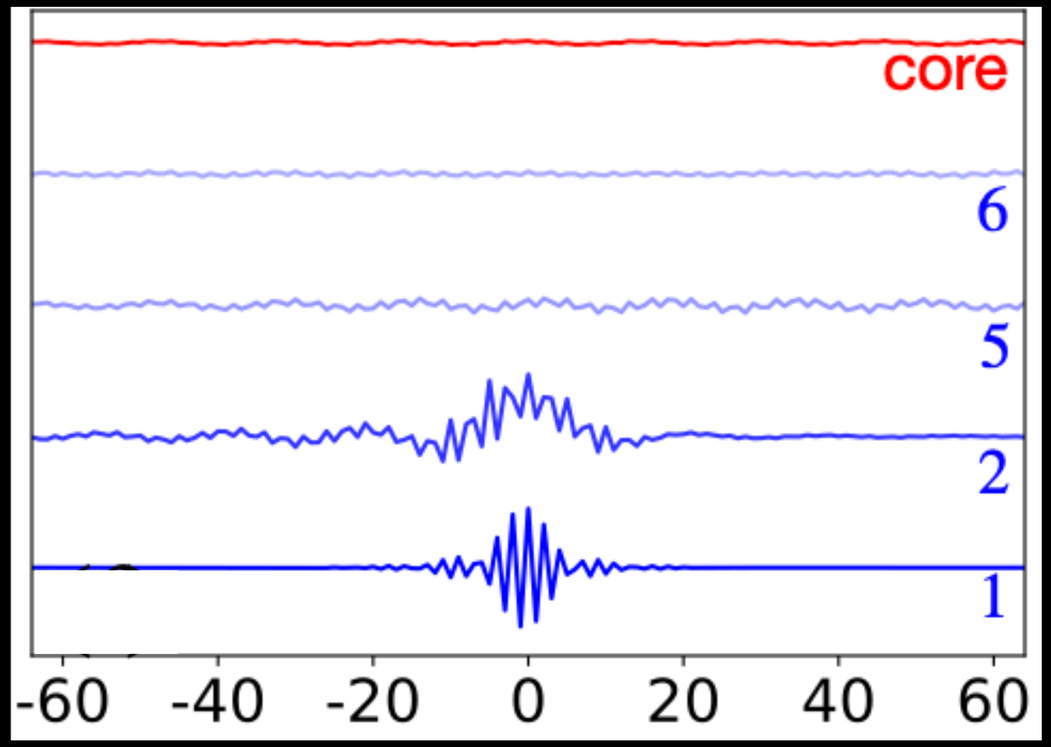


Nearest-Neighbor Model at half-filling

[Wong, Pang, & Po, 2206.11761]

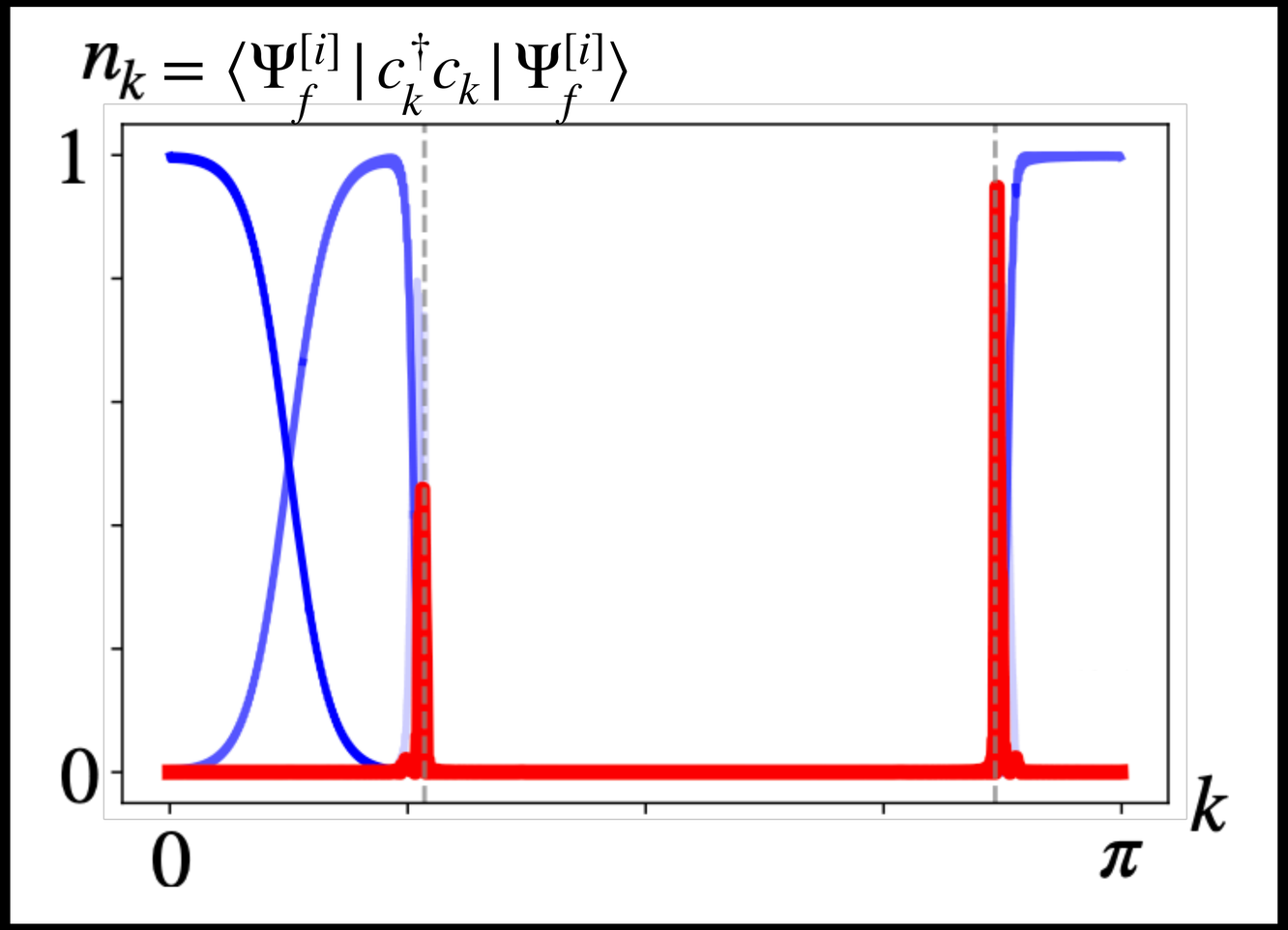
Extended Model - Next Nearest Neighbor Hopping and Electron Filling = 0.4





[Wong, Pang, & Po, 2206.11761]

Number occupation n_k at different RG times



(b) Extended Model with 0.4 filling

[Wong, Pang, & Po, 2206.11761]

Summary

Ansatz

Ansatz

Matrix Product States

$$S \sim \partial A \leq const$$

MERA

$$S \sim \log Vol(A)$$

Zipper Entanglement Renormalization

States

States

Q&A Session